RESEARCH PAPER

Society Under Monetary Stress II: The Cantillon Map

Asset Ownership, Monetary Expansion, and Wealth Stratification

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Abstract

This paper maps how U.S. households were differently positioned to experience asset-price and monetary-regime changes from 1989 through 2024. Using Federal Reserve Distributional Financial Accounts data and FRED macro series, it documents unequal portfolio exposure by wealth group. The top 10 percent held 60.9 percent of net worth in 1989 and 67.4 percent in 2024, while its equity share rose from 81.4 percent to 87.2 percent. The paper is descriptive, not causal: its contribution is a balance-sheet exposure map for interpreting household experience under macro stress.

JEL codes: D31, E21, E44, E51, G51 **Keywords:** household wealth, monetary expansion, asset ownership, distributional finance, balance sheets, wealth inequality **Disclosure:** This article is descriptive research using public data. It is not investment, legal, tax, or financial advice.

1. Introduction

Aggregate economic data often describe the country as if households occupy one representative balance sheet. That is convenient, but it hides the channel that matters most for ordinary economic life: households own different things. Some hold homes, equities, retirement accounts, and private business claims. Others hold wages, small cash buffers, consumer durables, and debt. When asset prices rise faster than wages, those two starting points produce very different lived experiences.

This paper calls that distributional starting point the Cantillon map. The term is used here in a practical rather than doctrinal sense. The central idea is simple: new purchasing power, easier credit, and rising asset prices do not reach every household in the same way or at the same time. The relevant question is not only whether nominal wealth rises in aggregate. It is who already owns the assets that reprice, who is still trying to buy into them from labor income, and who is carrying liabilities while entry prices move away.

The paper is the second in the *Society Under Monetary Stress* series. The first paper examined housing access and showed that the bridge from income to ownership became harder to cross even when monthly payment pressure moved through different interest-rate regimes. This paper generalizes the mechanism. Housing is one important asset. Equities, mutual funds, retirement balances, deposits, liabilities, and business claims are also part of the household map. A household with broad asset exposure experiences monetary and credit expansion differently from a household whose main asset is future labor income.

The contribution is descriptive. The paper does not claim that one monetary aggregate mechanically caused wealth inequality. It does not treat asset appreciation as artificial by definition. It also does not imply that every asset-owning household is secure or that every non-owner faces the same constraint. Instead, the paper uses public distributional data to organize a basic exposure map: macro regimes pass through portfolios. When portfolios are unequal, the same environment can strengthen balance sheets for one group while raising the ownership hurdle for another.

The paper is best read as a public data note and interpretive balance-sheet map, not as a causal monetary-policy paper. Its value is to make a known distributional reality legible for individual readers: asset ownership, cash/liquidity, wage dependence, and liability structure are different positions in the same macro environment.

That distinction matters for individual readers. A rising aggregate stock market, a rising home-price index, or a growing money stock can look like prosperity in broad charts. For a household trying to save from wages into ownership, the same environment can feel like a moving target. The useful question becomes: am I exposed to the assets being repriced, or am I trying to buy them later with income and cash that may not be keeping pace?

2. Relation to existing research

The idea that monetary and financial conditions have distributional effects is not new. Saez and Zucman document the long-run rise of U.S. wealth concentration and the changing position of the bottom 90 percent. Doepke and Schneider emphasize that inflation redistributes across households with different nominal asset and liability positions. Auclert formalizes redistribution channels of monetary policy, including earnings heterogeneity, Fisher effects, interest-rate exposure, and unhedged interest-rate exposure. Kaplan, Moll, and Violante show why household heterogeneity, liquid wealth, and marginal propensities to consume matter for monetary transmission. Coibion and coauthors study monetary policy and inequality using identified policy shocks, while Bartscher and coauthors examine monetary policy, asset prices, and racial wealth differences.

This paper does not attempt to replace that literature or estimate the same objects. Its narrower role is translational and descriptive. It uses the public DFA balance sheet to show the exposure surface that more formal work often models: who owns equities, real estate, deposits, and liabilities before asset prices and policy conditions move. The phrase “Cantillon map” is used as a metaphor for this exposure surface, not as a claim that the paper has identified a particular money-injection channel.

3. Data and measures

The main distributional source is the Federal Reserve's Distributional Financial Accounts. The DFA provides quarterly estimates of household wealth by wealth group, including the top 1 percent, the next 9 percent, the next 40 percent, and the bottom 50 percent. The estimates are constructed by integrating Financial Accounts aggregates with distributional information from the Survey of Consumer Finances and related allocation methods. They should therefore be read as official distributional estimates, not as direct census counts of every household balance sheet. The analysis combines the top 1 percent and next 9 percent into a top 10 percent group where useful, and compares that group with the bottom 90 percent and bottom half.

The sample runs from 1989 through 2024. For each year, the distributional balance-sheet observation uses the latest available quarter in that calendar year. The core balance-sheet variables are net worth, total assets, real estate, corporate equities and mutual fund shares, deposits, and liabilities. Shares are calculated from group-level dollar values, so the top 10 percent's equity share, for example, equals the top 1 percent plus next 9 percent's corporate equity and mutual fund holdings divided by aggregate household holdings of that category.

Public macroeconomic context comes from FRED series for M2, the consumer price index, the Case-Shiller national home-price index, median household income, average hourly earnings, the personal saving rate, and Federal Reserve assets. These series are not used as causal instruments. They provide context for the environment in which household balance sheets evolved.

Three indexed context measures are used cautiously. First, M2 divided by CPI gives a broad money-to-price-level context index. It is not a monetary shock, stance measure, or policy instrument. Second, the Case-Shiller home-price index divided by median household income gives a national home-price-to-income proxy. Third, aggregate DFA corporate equities and mutual fund shares divided by a real hourly wage proxy gives a rough balance-sheet scale measure: how large household-sector equity and mutual-fund claims became relative to an hourly wage benchmark. Each index is set to 1989 = 100. The equity-value-to-wage index is not a return series and should not be read as a clean causal measure of equities “beating wages.” It mixes valuation, retained earnings, retirement-system growth, portfolio allocation, household-sector scale, and labor-compensation trends. Its purpose is descriptive scale, not causal attribution.

The analysis is intentionally transparent. The goal is to show the map rather than estimate a fragile structural model. The descriptive design asks four questions:

1. How much net worth is held by different wealth groups? 2. How concentrated is equity ownership? 3. How did asset-value measures move relative to wages and income? 4. What does the bottom half's balance sheet look like when asset prices rise?

4. Results

4.1 Wealth shares shifted toward the top of the distribution

Table 1 summarizes the main distributional and macro indicators at selected milestones. In 1989, the top 10 percent held 60.9 percent of household net worth. By 2024, that share was 67.4 percent. The bottom 90 percent's share moved in the opposite direction, from 39.1 percent to 32.6 percent.

The change is not a straight line. The top 10 percent's share rose sharply through the housing boom and financialization years, reached 67.1 percent in 2007, and was 70.1 percent in 2019. It then eased to 67.4 percent by 2024, partly reflecting post-pandemic balance-sheet changes and asset-price dynamics across housing and financial markets. But the broad comparison remains clear: the ownership map in 2024 was more top-heavy than it was in 1989.

The bottom half's position is especially important for household interpretation. In 1989, the bottom 50 percent held 3.4 percent of net worth. In 2024, it held 2.5 percent. That is not a large enough balance-sheet share to participate meaningfully in aggregate asset appreciation. Even when the bottom half's nominal assets rise, its claim on aggregate net worth remains thin.

Table 1. Selected distributional and macro indicators

Year	Top 10% net-worth share	Bottom 90% net-worth share	Top 10% equity share	Bottom 90% equity share	Bottom 50% net-worth share	Bottom 50% equity share	M2/CPI index	Equity/wage index	Home-price/income index
1989	60.9	39.1	81.4	18.6	3.4	1.2	100	100	100
2000	62.6	37.4	79.8	20.2	3.2	1.5	113	406	96
2007	67.1	32.9	85.5	14.5	1.7	1.0	143	588	137
2019	70.1	29.9	88.3	11.7	1.7	0.6	236	1170	117
2024	67.4	32.6	87.2	12.8	2.5	1.0	273	1852	147

Source: Federal Reserve Distributional Financial Accounts and FRED public series. Indexes set 1989 = 100 where available.

4.2 Equity exposure was far more concentrated than population

The equity map is more concentrated than the net-worth map. In 1989, the top 10 percent held 81.4 percent of corporate equities and mutual fund shares. By 2024, it held 87.2 percent. The bottom 90 percent held 18.6 percent in 1989 and 12.8 percent in 2024.

The bottom half's equity exposure was tiny throughout the sample. It held 1.2 percent of corporate equities and mutual fund shares in 1989 and 1.0 percent in 2024. Its equity-to-assets ratio did rise over time, from 1.5 percent of assets in 1989 to 5.1 percent in 2024, but from such a low base that it did not materially change the aggregate ownership map.

This matters because equity appreciation is one of the main channels through which expansionary financial conditions can raise household net worth. If most equity claims are already held by the top decile, then a large rise in equity values mechanically benefits that group more in dollar terms. This does not require a moral claim about the owners. It is simply portfolio arithmetic.

The top 1 percent shows the point sharply. In 1989, corporate equities and mutual fund shares were 18.7 percent of top-1-percent assets. By 2024, they were 47.3 percent. The group was not merely rich in level terms. It became much more directly exposed to the asset class that expanded most dramatically in household balance sheets.

4.3 Asset-value scale measures rose faster than ordinary income benchmarks

The macro context measures show why the ownership map matters. The M2/CPI index rose from 100 in 1989 to 273 in 2024. The home-price-to-income proxy rose from 100 to 147. The aggregate corporate-equity-value-to-real-wage-proxy index rose from 100 to 1,852.

These measures should be read carefully. The M2/CPI index is not a causal variable that explains everything else. The equity-value-to-wage index is not an investable return series. The home-price-to-income index is a national proxy and does not capture down-payment constraints, credit standards, local variation, taxes, or insurance. Still, the direction is hard to ignore. The balance-sheet value of major assets grew much faster than ordinary labor-income benchmarks.

For households that already owned those assets, this environment could strengthen net worth even if wages felt ordinary. For households trying to accumulate from wages and cash, the same environment increased the scale of the asset hurdle. A household does not buy a diversified balance sheet with an index; it buys shares, retirement contributions, a down payment, or a home at prevailing prices. If those prices move faster than income, delayed entry becomes costly.

This is the practical Cantillon map. The question is not only whether the economy grew. It is whether the household was positioned before the repricing. Ownership timing becomes an important part of household exposure.

4.4 The bottom half remained balance-sheet fragile

The bottom half's balance sheet is not simply a smaller version of the top decile's balance sheet. In 2024, liabilities were 59.7 percent of bottom-half assets, close to the 58.8 percent level observed in 1989. Real estate was 36.6 percent of assets, deposits were 10.4 percent, and corporate equities and mutual fund shares were 5.1 percent.

That combination is fragile. The bottom half has some asset exposure, but not enough to receive much of aggregate equity appreciation. It has housing exposure, but often paired with leverage and local affordability risk. It has deposits, but deposits are more protective as liquidity than as long-run participation in asset appreciation. And it carries liabilities large enough to make interest-rate and income shocks matter.

Table 2. Balance-sheet composition by wealth group, 2024

Wealth group	Equity / assets	Real estate / assets	Deposits / assets	Liabilities / assets	Net-worth share
Top 1%	47.3	12.3	6.4	2.0	31.0
Next 9%	28.8	22.7	7.8	6.3	36.4
Next 40%	10.0	38.0	8.7	15.0	30.1
Bottom 50%	5.1	36.6	10.4	59.7	2.5

Source: Federal Reserve Distributional Financial Accounts. Entries are percentages.

This balance-sheet structure helps explain why aggregate wealth gains can coexist with financial stress. If asset prices rise, a household with broad claims may gain. A household with limited claims may see the same rise as a higher entry price. If interest rates rise, a leveraged household may face tighter cash flow even if nominal asset values remain high. If wages lag behind the asset ladder, saving becomes a slower bridge to ownership.

The bottom half therefore faces a double challenge: low participation in the fastest-appreciating balance-sheet categories and persistent vulnerability to liabilities. That does not mean households are helpless. It means the relevant personal-finance problem is not merely budgeting. It is gaining durable exposure to productive and scarce assets while preserving liquidity and avoiding debt structures that turn volatility into forced selling or delayed accumulation.

5. Reading the evidence as a household

The household lesson is not that everyone should own the same portfolio. Households differ in age, health, family obligations, career stability, location, and risk capacity. The lesson is more basic: know which side of the repricing you are on.

A household can use the Cantillon map as a diagnostic checklist.

First, compare wage growth with target-asset prices. If income is rising but home prices, equity prices, tuition, insurance, or other ownership hurdles are rising faster, nominal progress may still leave the household behind.

Second, distinguish liquidity from long-run exposure. Deposits and emergency savings are essential. They reduce fragility and protect decision-making. But cash-like balances do not provide the same exposure as ownership of productive or scarce assets. A household that holds only liquidity may feel safe in the short run while falling behind the asset ladder over longer periods.

Third, watch leverage quality. Debt is not one thing. Fixed-rate, serviceable, productive debt differs from variable-rate, consumption-driven, or fragile debt. The bottom-half balance sheet shows why liabilities matter: when debt is large relative to assets, macro volatility can quickly become household stress.

Fourth, separate monthly affordability from entry affordability. A low rate can make payments look manageable while the down-payment hurdle rises. A high rate can compress demand while still leaving prices elevated. The ownership problem is the combined burden of price, income, cash entry, financing terms, and local market conditions.

Fifth, think in terms of sequence. Households already holding assets before a repricing event experience that event differently from households trying to buy afterward. This is not a reason for panic. It is a reason to build disciplined, resilient exposure over time rather than treating accumulation as something to begin only after conditions feel easy.

6. Limits, measurement, and interpretation

The analysis has important limits. It is descriptive, national, and balance-sheet focused. It does not identify a single causal shock. It does not separate every channel through which taxes, technology, globalization, housing supply, demographics, market concentration, public policy, financial innovation, retirement-account expansion, saving-rate differences, and business ownership affect wealth shares. It also does not observe every household's full financial situation or risk tolerance.

The DFA is a powerful public source, but it is still an aggregate distributional system. Wealth groups are not fixed households across time. A household can move between groups. Asset categories also hide important variation. A primary residence, rental property, concentrated stock position, broad retirement fund, private business, and cash deposit all behave differently in practice.

The macro indexes are context measures, not structural estimates. M2/CPI can rise for reasons that are entangled with crisis response, banking conditions, policy choices, and private credit behavior. Home-price-to-income pressure varies sharply by region. Equity-value-to-wage pressure partly reflects retained earnings, valuation multiples, household allocation, retirement-system design, and broader financial deepening.

These limits narrow the claims, but they do not erase the map. The robust descriptive fact is that households were differently exposed. The top decile held most net worth and an even larger share of equity claims. The bottom half held a small share of net worth, a tiny share of equity claims, and substantial liabilities relative to assets. In a world where asset values rose faster than labor benchmarks, those starting positions mattered.

7. Conclusion

The modern household economy cannot be understood only through aggregate growth, inflation, or income statistics. It has to be read through balance sheets. From 1989 through 2024, U.S. households entered monetary and credit regimes with sharply different portfolios. The top 10 percent held most net worth and the overwhelming majority of corporate equities and mutual fund shares. The bottom half held very little net worth, almost no aggregate equity exposure, and liabilities that remained large relative to assets.

That is the Cantillon map. It shows why nominal expansion can feel like prosperity from one balance sheet and pressure from another. Asset owners participate directly in repricing. Wage-dependent and cash-dependent households must save into repriced assets after the fact. Leveraged households may gain from inflation in some scenarios but can be squeezed by rate resets, income shocks, or elevated entry prices.

The individual implication is practical. Households should not read macro conditions only as headlines. They should ask how their own balance sheet is positioned: income growth, emergency liquidity, asset exposure, housing-entry hurdle, debt structure, and time horizon. A sounder personal strategy begins with seeing the map clearly. The public data suggest that ownership exposure, timing, and resilience are not side issues. They are central to how monetary stress becomes household reality.

Replication note

The distributional panel uses the Federal Reserve DFA public CSV bundle, specifically the net-worth levels detail and net-worth shares files. Annual observations use the latest available DFA quarter in each calendar year. Top 10 percent values combine the top 0.1 percent, remaining top 1 percent, and next 9 percent categories as reported in the DFA files; in the downloaded file structure used here, the top 1 percent is assembled from the top 0.1 percent and remaining top 1 percent rows. FRED context series are annualized from public graph CSV endpoints. Monthly, weekly, or daily series are averaged within calendar year; annual series use their annual value. Constructed indexes are normalized to 1989 = 100.

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